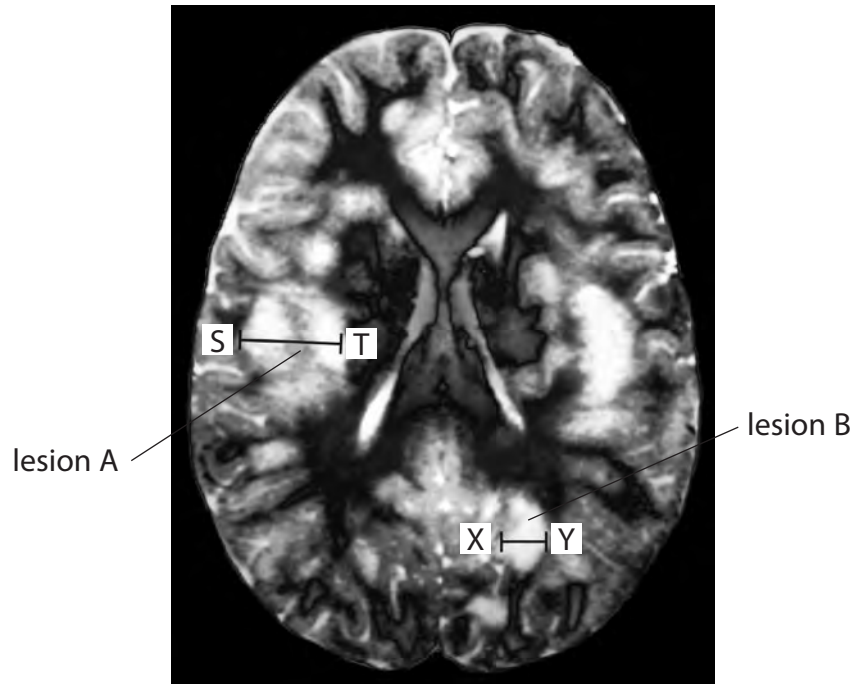


- 3 Acute disseminated encephalomyelitis (ADEM) causes the destruction of the myelin sheath around neurones.

The destruction is the result of inflammation.

Lesions form in the brain where the myelin is destroyed.

The image is a magnetic resonance imaging (MRI) scan of a person with ADEM, showing lesions in the brain.



(Source: © ZEPHYR/SCIENCE PHOTO LIBRARY)

- (a) (i) Lesion A is spherical and has a diameter (S–T) of 1.97 cm.

Calculate the volume of lesion A.

Use the formula:

$$V = \frac{4}{3}\pi r^3$$

Give your answer to 2 decimal places.

(2)

Answer cm³

- (ii) Calculate the actual width of the lesion labelled B between points X and Y.
Give your answer to 2 significant figures.

(2)

Answer cm

- (b) An MRI scan can be used to diagnose a brain tumour.

Suggest why a dye might be given during the MRI scan.

(1)

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- (c) A person with ADEM has problems with walking.

Explain why ADEM causes problems with walking.

(2)

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(Total for Question 3 = 7 marks)